

Phylogenetic Analysis of the Hemoglobin Beta (HBB) Gene

Introduction

The **hemoglobin beta (HBB) gene** encodes the β -globin polypeptide, one of the two major globin chains that form adult hemoglobin (HbA), the oxygen-carrying protein in the red blood cells of humans and other mammals. Hemoglobin is responsible for the transport of oxygen from the lungs to peripheral tissues and facilitates the return transport of carbon dioxide and protons from tissues to the lungs, hence allowing acid–base homeostasis to take place. The β -globin chain is integral for the structural integrity and physiological function of hemoglobin, making the HBB gene one of the most extensively studied genes in molecular biology, genetics, and evolutionary biology.

In humans, the **HBB gene is located on chromosome 11p15.5** within the β -globin gene cluster, which also contains the embryonic (HBE1), fetal (HBG1 and HBG2), and minor adult (HBD) globin genes. This cluster has evolved through gene duplication events that allow expression during development, only specific to a certain stage. The HBB gene encodes a **147-amino acid β -globin protein** with a molecular mass of approximately **15.9 kDa**. Following translation, two α -globin and two β -globin chains assemble with four heme prosthetic groups to form the mature hemoglobin tetramer.

Unlike many proteins, β -globin does not contain multiple independently folded domains; rather, the entire polypeptide functions as a single globin domain. Within this domain lies a hydrophobic **heme-binding pocket**. Additional conserved residues enable tetramer assembly and allow cooperative oxygen binding.

Mutations in the HBB gene can disrupt the structure, stability, or function of the β -globin protein, leading to a multitude of hemoglobin disorders, known as **hemoglobinopathies**. One of the best-characterized examples is sickle cell anemia, in which a single nucleotide substitution replaces glutamic acid with valine at position 6 of the beta globin chain. This substitution causes erythrocyte sickling and the clinical manifestations of sickle cell disease. Other mutations may bring changes in oxygen affinity, destabilize the tetramer, resulting in unstable hemoglobin variants. In contrast, mutations affecting HBB transcription, RNA processing, or translation reduce β -globin synthesis and give rise to **β -thalassemia**, characterized by ineffective erythropoiesis (RBC production), and chronic anemia. The severity of these disorders depends on the specific mutation and its effect on β -globin production or function.

Phylogenetic analyses of HBB sequences consistently demonstrate that the gene is **highly conserved across mammals**, showing that its structure and function has remained preserved. Mammalian HBB genes show generally high nucleotide and amino acid sequence identity. The conserved coding regions maintain critical amino acid residues that cause **heme binding, $\alpha\beta$ globin interactions, and oxygen binding**. On the other hand, amino acid changes in certain mammalian lineages inhabiting extreme environments have been identified, such as hypoxic

habitats (low oxygen), where modifications in oxygen affinity increase physiological performance without disturbing overall protein stability.

In this project, the human HBB protein sequence (**RefSeq Accession: NP_000509.1**) serves as the reference sequence for identifying homologous HBB proteins from multiple mammalian species using **BLASTp**. The retrieved protein sequences are aligned using the **ClustalW** algorithm in **MEGA 12** to identify conserved amino acid residues and functionally important regions. The aligned sequences are subsequently used to construct a **Neighbor-Joining phylogenetic tree**, providing insights into the evolutionary relationships among the HBB proteins across different species. The resulting phylogenetic tree is further visualized and annotated using **Interactive Tree of Life (iTOL)** to improve interpretation and presentation.

Data Retrieval and Phylogenetic Analysis

In this project, the human HBB protein (**RefSeq Accession: NP_000509.1**) was selected as the reference (query) sequence for comparative analysis. Proteins evolve more slowly than nucleotide sequences due to the functional constraints imposed on amino acid composition and protein structure, hence a protein-based phylogenetic approach is a reliable framework for studying evolutionary relationships among species.

Methodology

The human HBB protein sequence (**RefSeq Accession: NP_000509.1**) was retrieved from the **NCBI protein database** and used as the reference sequence for this project. A **BLASTp** search was performed against the NCBI protein database to identify homologous HBB protein from diverse **mammalian species (25 species)**. The top homologous protein sequences were selected based on **sequence similarity, query coverage, and E-value, and downloaded in FASTA format**. Proteins exhibiting high sequence identity, query coverage and extremely low E-values are considered strong homologous candidates and are therefore suitable for comparative evolutionary analysis.

The retrieved protein sequences, together with the human HBB query sequence, were imported into **MEGA 12**, where a **multiple sequence alignment (MSA)** was performed using the **ClustalW** algorithm. The alignment allowed direct comparison between the reference protein and its homologs, letting us identify the conserved amino acid residues, variable regions and evolutionary divergence among species. The aligned sequences were subsequently used to construct a **Neighbor-Joining (NJ) phylogenetic tree** with **1,000 bootstrap** replicates to assess the reliability of the inferred evolutionary relationships. Finally, the phylogenetic tree was exported in the **Newick (.nwk)** format and visualized using the **Interactive Tree of Life (iTOL)** platform for annotation and presentation.

Tools Used

Tool/Database	Purpose
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NCBI Protein	Retrieval of human HBB protein sequence (NP_000509.1).
BLASTp	Identification of homologous HBB protein sequences based on sequence similarity.
MEGA 12	Multiple sequence alignment (ClustalW) and phylogenetic tree construction using the Neighbor-Joining method.
iTOL (Interactive Tree of Life)	Visualization, annotation, and customization of the phylogenetic tree for interpretation and presentation.